

Dev Jariwala

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EDUCATION

University of Michigan

B.S.E. in Computer Science

Ann Arbor, MI

May 2028

Relevant Coursework: Data Structures and Algorithms, Computer Architecture, Object-Oriented Programming

EXPERIENCE

Hyundai America Technical Center (HATCI) via KPIT Technologies

Novi, MI

Software Engineer Intern, Client Assignment

July 2026 – Aug. 2026

- Built **20+ automated Robot Framework test cases** in **Python**, covering **5 Hyundai ccNC modules**.
- Expanded automated regression coverage to **SiriusXM 360L** infotainment services across vehicle trims.
- Investigated infotainment defects across **5 modules** alongside a **5-person KPIT engineering team**.
- Verified release fixes through manual in-vehicle testing within a **100+ engineer** HATCI organization.

KPIT Technologies

Novi, MI

Software Engineer Intern

June 2026 – July 2026

- Built a **JWT-based** authentication platform in **Angular**, **Flask**, and **MySQL** for internal tooling.
- Implemented granular role-based access control for **50+ internal users** across platform modules.
- Developed **6 REST APIs** across **10+ frontend pages**, reducing manual user provisioning steps by **40%**.

Necessities

Ann Arbor, MI

Software Engineer Intern

Sep. 2025 – Apr. 2026

- Built **50+ reusable React components** for marketplace search, filtering, and discovery.
- Built **PostgreSQL-backed** REST APIs supporting messaging for **100+ active marketplace users**.
- Reduced page load times by **30%** across **1,000+ listings** through targeted PostgreSQL query tuning.
- Delivered biweekly releases within a **5-person engineering team** using CI/CD workflows.

PROJECTS

StormRoute | [GitHub](#) · [Live Demo](#) | *Next.js, TypeScript, React, Leaflet, Open-Meteo, OSRM, Vitest*

- Built a weather-aware routing engine in **Next.js/TypeScript** evaluating **6 departure windows**.
- Matched hourly weather forecasts with road geometry using **Leaflet** and the **OSRM** routing engine.
- Scored route risk on a deterministic **0–100 scale** across **6 weather components**, comparing **3 routes**.
- Reduced route-analysis latency by **99.91%** through warm-cache reuse with zero repeated weather calls.
- Tested the core pipeline with **82 automated tests** in **Vitest**, reaching **98.5% line** and **94.0% branch coverage**.

MiniRedis | [GitHub](#) | *C++20, TCP/IP, Multithreading, CMake*

- Built a Redis-style key-value store in **C++20** supporting **6 core Redis commands** over TCP connections.
- Implemented thread-safe storage using reader-writer locks and a fixed-size multithreaded worker pool.
- Added crash recovery with **TTL expiration**, **LRU eviction**, and append-only persistence logging.
- Reached **52,000+ operations/sec** across **25,000** benchmarked requests under concurrent client load.

Cricket Scoring & Analytics Platform | *React, TypeScript, Supabase, PostgreSQL, GCP*

- Built a cricket scoring platform supporting **3 synchronized workflows** through WebSockets.
- Implemented a scoring engine covering **15+ tournament rules**, including dismissals and extras.
- Automated second-innings target calculations and live player statistics updates.
- Protected application data through **3 authorization layers**: OAuth 2.0, row-level security, and RBAC.

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, TypeScript, SQL

Frameworks/Libraries: React, Angular, Flask, Robot Framework, Flutter

Databases: PostgreSQL, MySQL, Firestore, Supabase

Cloud/Developer Tools: Google Cloud Platform (GCP), Docker, Git, Linux, Vercel, Jumpstarter, CMake, Ollama